

BLUE TEAM LOG ANALYSIS

Understanding, Analyzing & Responding to Logs in the Real World

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WHAT IS A LOG?

A log is a timestamped record of everything that happens on a system — logins, file opens, network connections, errors. Think of it as the computer's diary. Every time something happens, it writes it down.



Detect Attacks

Spot brute force, malware running, stolen accounts



Investigate Incidents

Reconstruct exactly what an attacker did, step by step



Respond Fast

Cut dwell time — the time attacker stays hidden



Prove What Happened

Logs are evidence — in court and in incident reports

■ REAL WORLD — SolarWinds (2020)

APT29 (Cozy Bear — Russian state hackers) hid inside SolarWinds Orion software for **9 MONTHS** before anyone noticed. How? **No one was checking the logs.** The attack infected 18,000 organizations including US government agencies. If defenders had been reading their network & authentication logs, they would have seen strange traffic to unknown domains within days — not months.

LOG TYPES — THE 8 DIARIES YOUR SYSTEM KEEPS

SYS

System Log

Your OS (Windows, Linux, macOS) keeps a diary of everything that happens inside it.

e.g. Windows Event Log, /var/log/syslog, journald

NET

Network Log

Every time your computer sends or receives data, a record is created.

e.g. NetFlow/IPFIX, firewall logs, IDS/IPS alerts, packet captures

APP

Application Log

Apps like websites, databases, and APIs generate their own logs whenever someone uses them.

e.g. Apache/Nginx, SQL query logs, API access logs, crash reports

SEC

Security Log

Security tools like antivirus, EDR, and firewalls generate alerts when they spot something suspicious.

e.g. AV/EDR alerts, WAF events, DLP violations, SIEM rules

AUTH

Authentication Log

Every time someone logs in or out — a website, an app, a VPN — an auth log entry is created.

e.g. Windows Event 4624/4625, SSH auth.log, MFA attempts, VPN sessions

DNS

DNS Log

DNS is like the internet's phone book. Every lookup for a domain creates a log entry.

e.g. Domain lookups, NXDOMAIN replies, DNS tunneling, PTR queries

CLD

Cloud Log

Every action taken in cloud services (AWS, Azure, GCP) is logged.

e.g. AWS CloudTrail, Azure Monitor, GCP Audit Logs, S3 access logs

EML

Email Log

Email is the #1 way attackers get in. Every email sent and received generates a log entry.

e.g. SMTP headers, SPF/DKIM/DMARC, attachment hashes, click-through URLs

THE 7-STEP LOG ANALYSIS METHOD

01

COLLECT & CENTRALISE

Pull logs from ALL sources into one SIEM (Splunk/ELK). Normalize time to UTC.

02

PARSE & NORMALISE

Break raw text into fields: timestamp | user | src_ip | action | result

03

ESTABLISH A BASELINE

Learn what NORMAL looks like first. You can't spot weird without knowing normal.

04

APPLY DETECTION LOGIC

Run rules & thresholds: '20 failed logins in 5 min = alert'. Use Sigma rules.

05

CONTEXTUALISE & CORRELATE

Never read one log line alone. Join auth + network + DNS to build the full story.

06

ANSWER THE 5W1H

What? When? Where? Who? How? Why? — every unanswered question is a gap.

07

DETERMINE DISPOSITION

True Positive / False Positive / Benign True Positive.

TOOLS EVERY BLUE TEAMER USES

SIEM

Splunk

Splunk is the industry-standard platform for collecting, searching, and analyzing logs. It pulls logs from every system in your organization into one place, so you can search them all at once using a simple query language called **SPL**. Think of it as **Google for your logs** — you type what you're looking for and Splunk finds it across millions of log lines in seconds. It's used in SOC teams worldwide to detect attacks, investigate incidents, and build real-time dashboards.

ANALYZING LOGS IN THE TERMINAL

1■■ See all failed login attempts

```
grep "Failed password" training.log
```

3■■ See all successful logins

```
grep "Accepted" training.log
```

4■■ See all cron/scheduled tasks

```
grep -i cron training.log
```

5■■ See all EDR alerts

```
grep -i alert training.log
```

6■■ See all blocked connections

```
grep -i blocked training.log
```

7■■ See all DNS queries

```
grep -i dns training.log
```

■ Search all log files in a directory

```
grep -R "Failed password" .
```

THE 5W1H INVESTIGATION FRAMEWORK

Every unanswered question is an investigation gap. Use this to document incidents.

WHAT Event type, affected asset, action taken, artifacts left behind <i>Example: Ransomware (LockBit 3.0) encrypted 3,400 files on finance server FS-01</i>	WHEN Exact timestamp, duration, how long attacker was hidden (dwell time) <i>Example: Initial click 08:14 → encryption 08:22 → detected 09:05. Dwell time: 51 min</i>	WHERE Source system, destination, how it spread across the network <i>Example: Started on WS-ACCT-04 → spread to FS-FINANCE-01 via SMB protocol</i>
WHO User account involved, which process, real person or automated script <i>Example: amara.bekele@company.com — account compromised via phishing 45 min earlier</i>	HOW Attack method, tools used, MITRE ATT&CK; technique, evasion tactics <i>Example: Phishing → macro → PowerShell → NTLM hash stolen → LockBit deployed</i>	WHY Motive, root cause, what gap let the attacker succeed <i>Example: Financial motive. Root cause: no MFA + unpatched EternalBlue (CVE-2017-0144)</i>

DETERMINE DISPOSITION

Every alert must get a final verdict. 'I didn't decide' is NOT an option.

■ TRUE POSITIVE (TP)

When: The alert is REAL. An actual attack happened.

Action: Escalate by severity. Contain the host. Investigate. Open an IR ticket. Notify management.

EXAMPLE: 15 failed logins → 1 success → attacker downloaded 4GB. REAL attack.

■ FALSE POSITIVE (FP)

When: The alert fired but there is NO real threat.

Action: Tune the detection rule so it doesn't fire on this normal activity again. Document why.

EXAMPLE: IT team ran a scheduled script that triggered the brute-force rule. Not an attack.

■ BENIGN TRUE POSITIVE (BTP)

When: The behaviour happened BUT it is not malicious.

Action: Close with full documentation explaining why it's benign. Still write down your reasoning.

EXAMPLE: Developer ran a port scan on the TEST network — technically triggered IDS but was authorized.

■■ 40% false positive rate = analysts dismiss half their alerts. This causes ALERT FATIGUE — real attacks slip through. TUNE your rules.

REAL WORLD — FULL ATTACK CAUGHT: INSIDER THREAT

Scenario: Ethiopian Telecom employee **solomon.tadesse** logged in at 10PM, copied 4,200 files (2.1GB) to Dropbox and disappeared. Detected by the SIEM.

1	COLLECT All auth & file-access logs from 800 workstations pulled into Splunk. UTC timestamps normalised.
2	BASELINE Normal: solomon logs in 07:30–17:00, ~200 files/day, <50MB. Never touches HR salary database.
3	DETECT Alert at 22:15 — solomon accessed 4,200 files, transferred 2.1GB in one session. 42× above baseline.
4	CORRELATE Timeline: login 22:10 → HR salary DB accessed → bulk copy to C:\Temp → Dropbox upload → logout 23:45
5	5W1H WHAT: Data exfil WHEN: 22:10–23:45 WHO: solomon HOW: legit creds + browser WHY: competing job offer
6	DISPOSITION TRUE POSITIVE. Account disabled. Legal + HR notified. DLP rule updated to block bulk uploads to personal cloud.

■ WannaCry (2017) — 230,000 machines in 150 countries in ONE DAY. NHS UK lost £92M. Root cause: no patching playbook (MS17-010 had been out 2 months). No offline backups. Blue teams that had prep playbooks were safe in hours.

REAL WORLD — LAZARUS GROUP IN ACTION

North Korean state hackers. Stole \$81M from Bangladesh Bank (2016) and hacked Sony Pictures (2014).
How do logs catch them?

INITIAL ACCESS EMAIL LOG	Attack: Spear-phishing email impersonating LinkedIn job recruiters with malicious PDF DETECTED: Email log shows: sender domain spoofed, PDF attachment flagged by scanner, link in body
EXECUTION SYSTEM LOG	Attack: Victim opens PDF → macro runs → PowerShell downloads dropper malware DETECTED: Sysmon shows: acrobat.exe spawned powershell.exe → downloaded unknown binary
C2 COMMUNICATION DNS LOG	Attack: Malware contacts attacker server: lazarusc2.xyz, update-secure.net DETECTED: DNS log shows lookup of lazarusc2.xyz — check VirusTotal → flagged as malicious
LATERAL MOVEMENT AUTH LOG	Attack: Stolen credentials used to move from workstation to SWIFT banking server DETECTED: Auth log: account logged into SWIFT server at 3AM from new machine — never seen before
EXFILTRATION NETWORK LOG	Attack: \$81M transferred via SWIFT. Data sent to attacker IPs over uncommon ports DETECTED: Large bytes transferred to foreign IP on port 8443 — not standard SWIFT traffic

LOG PATHS — WHERE TO FIND THEM

LOG TYPE	LINUX PATH	WINDOWS SOURCE
Authentication	sudo journalctl -n 50	Event ID 4624/4625 (Security Log)
System	/var/log/syslog or /var/log/messages	Event ID 7045 (System Log)
DNS	/var/log/syslog (bind9) or /var/log/named/	DNS Debug Logs
Network/Firewall	/var/log/ufw.log or iptables log	Windows Firewall Log
Web Server (Nginx)	/var/log/nginx/access.log & error.log	IIS: C:\inetpub\logs
Web Server (Apache)	/var/log/apache2/access.log	N/A
Cron / Scheduled Tasks	/var/spool/cron/ & /etc/cron*	Task Scheduler Log
Email (Postfix)	/var/log/mail.log	Exchange Logs

■ Pro tip: On Kali/Linux, run `sudo find / -name '*.log' 2>/dev/null` to discover ALL log files on the system.

TEST YOUR KNOWLEDGE

01

What does a large number of failed login attempts in logs usually indicate?

■ *Hint: Think about what an attacker does when they don't know a password...*

02

Why is it suspicious if a computer suddenly starts connecting to many unknown IP addresses?

■ *Hint: What does malware need to do after it infects a machine?*

03

Why do analysts check logs?

■ *Hint: What are the 4 main things logs help you do as a Blue Teamer?*